

Homework 1 - EE4520 Analog CMOS Design 1

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1 Simulation 1: Biasing Circuit

1.1 Objective

Design a biasing circuit with six MOSFETs (Mp1, Mp2, Mp3, Mn1, Mn2, Mn3) to achieve a main bias current $I_{b,main} = 200 \mu\text{A}$ with overdrive voltages $V_{gt,n1} \approx V_{gt,p1} \approx 100 \text{ mV}$. Determine the ratios $ratio_n$ and $ratio_p$ to keep Mn1 and Mp1 at the edge of saturation with approximately 50 mV margin.

1.2 Design Parameters

All transistors use fixed dimensions:

- $W = 1.00 \mu\text{m}$
- $L = 0.18 \mu\text{m}$

The M-factors were chosen to achieve $V_{gt} \approx 100 \text{ mV}$:

- $M_n = 11$ (for Mn1, Mn2, Mn3)
- $M_p = 60$ (for Mp1, Mp2, Mp3)

1.3 Determination of Ratio_n

To find $ratio_n$, a DC sweep was performed on R_n from 3 to 9 with step 0.1, monitoring the drain-source voltage of Mn1 (node V_{sn}). The target is $V_{ds,Mn1} = V_{dsat,Mn1} + 50 \text{ mV}$ to ensure Mn1 remains in saturation with adequate margin.

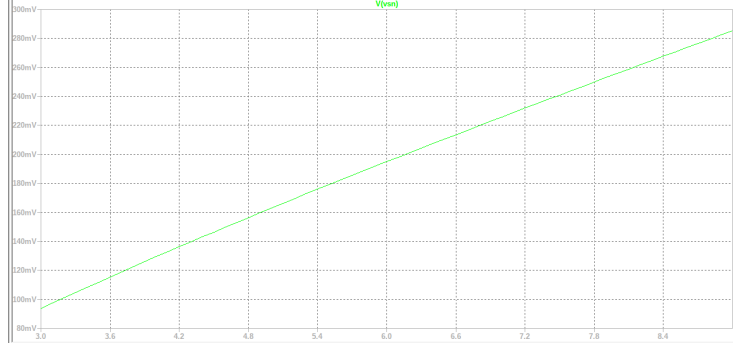


Figure 1: $V(V_{sn})$ vs R_n sweep showing Mn1 drain voltage as a function of the NMOS current ratio.

From the sweep, at $R_n = 5.8$, $V_{sn} \approx 189$ mV, which provides approximately 50-60 mV margin above the saturation voltage. Therefore:

$$\boxed{ratio_n = 5.8} \quad (1)$$

1.4 Determination of Ratio_p

Similarly, a DC sweep was performed on R_p from 3 to 12 with step 0.1, monitoring node V_{sp} . Since Mp1 is a PMOS, $|V_{ds,Mp1}| = V_{dd} - V_{sp}$. The target is $|V_{ds,Mp1}| = |V_{dsat,Mp1}| + 50$ mV.

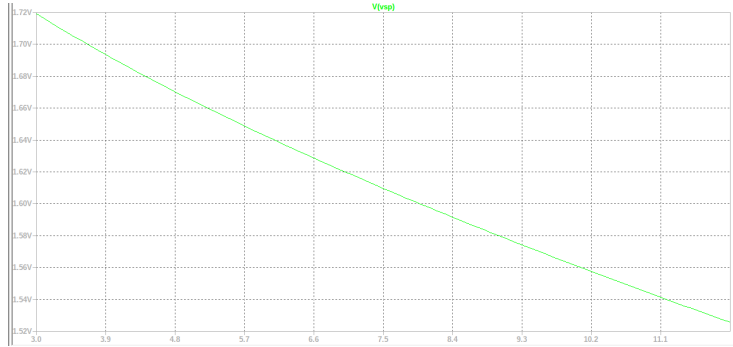


Figure 2: $V(V_{sp})$ vs R_p sweep showing Mp1 source voltage as a function of the PMOS current ratio.

From the sweep, at $R_p = 7.5$, $V_{sp} \approx 1.61$ V, giving $|V_{ds,Mp1}| \approx 190$ mV with adequate saturation margin. Therefore:

$$\boxed{ratio_p = 7.5} \quad (2)$$

1.5 Why Ratios Differ from Theoretical Value of 4

Theoretically, to achieve $V_{cn} = 2 \times V_{gs}$ for the cascode bias, we expect $ratio_n = ratio_p = 4$. However, the actual ratios are significantly higher due to:

1. **Body Effect:** Mn2 has its source above ground (at V_{sn}), causing its threshold voltage to increase. This requires higher $V_{gs,Mn2}$ to maintain the same current, shifting the voltage division and requiring larger $I_{b,ncas}$.
2. **Channel Length Modulation:** The finite output resistance of short-channel transistors causes the drain current to depend slightly on V_{ds} , affecting the voltage distribution.
3. **BSIM3v3 Model Effects:** The advanced transistor model includes mobility degradation, velocity saturation, and other short-channel effects not captured in simple square-law equations.
4. **PMOS vs NMOS Asymmetry:** The PMOS transistors have lower mobility than NMOS, and Mp2 experiences body effect differently, leading to $ratio_p > ratio_n$.

1.6 Circuit Diagram with Node Voltages

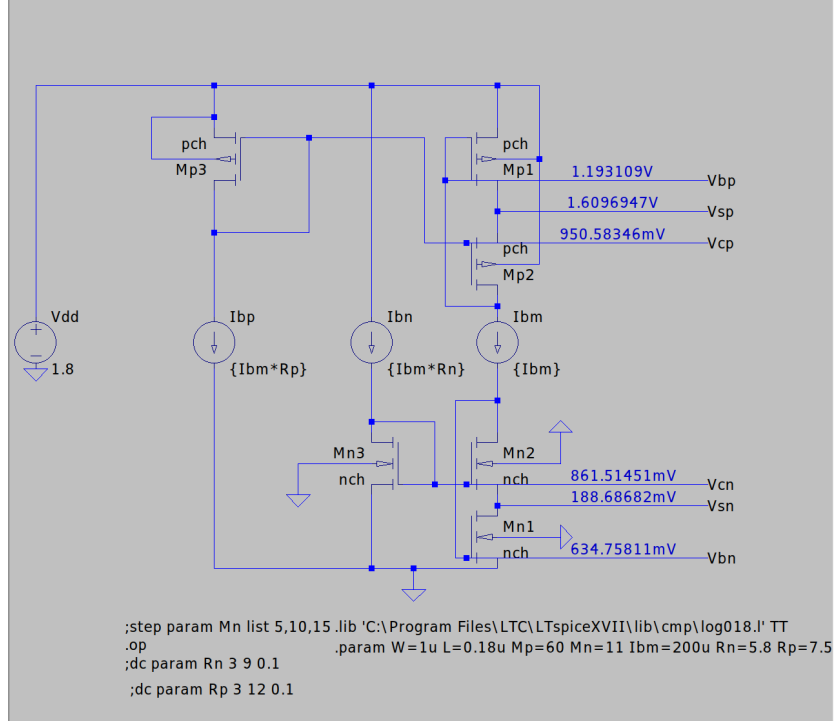


Figure 3: Biasing circuit schematic with annotated node voltages at $I_{b,main} = 200 \mu\text{A}$.

1.7 Operating Point Results

Table 1 summarizes the node voltages and branch currents for three bias current conditions: $200 \mu\text{A}$ (nominal), $100 \mu\text{A}$ ($2\times$ reduction), and $50 \mu\text{A}$ ($4\times$ reduction).

Table 1: Node voltages for different bias current levels			
Node	$I_{b,main} = 200 \mu\text{A}$	$I_{b,main} = 100 \mu\text{A}$	$I_{b,main} = 50 \mu\text{A}$
V_{bn} (mV)	634.8	583.9	541.3
V_{sn} (mV)	188.7	128.6	97.3
V_{cn} (mV)	861.5	732.0	649.0
V_{cp} (mV)	950.6	1069.5	1152.0
V_{sp} (mV)	1609.7	1661.7	1692.5
V_{bp} (mV)	1193.1	1242.6	1284.3

Table 2: Branch currents for different bias current levels

Current	$I_{b,main} = 200 \mu\text{A}$	$I_{b,main} = 100 \mu\text{A}$	$I_{b,main} = 50 \mu\text{A}$
$I_{b,main} (\mu\text{A})$	200	100	50
$I_{b,ncas} (\mu\text{A})$	1160	580	290
$I_{b,pcas} (\mu\text{A})$	1500	750	375
$I_{d,Mn1} (\mu\text{A})$	200	100	50
$I_{d,Mp1} (\mu\text{A})$	200	100	50

1.8 Observations on Current Scaling

When the bias current is reduced by $2\times$ and $4\times$:

- All node voltages shift accordingly due to the square-law relationship between current and overdrive voltage ($I_d \propto V_{gt}^2$).
- V_{sn} decreases from 189 mV to 97 mV, indicating reduced V_{ds} margin for Mn1 at lower currents.
- V_{sp} increases toward V_{dd} , similarly reducing the $|V_{ds}|$ margin for Mp1.
- The cascode bias voltages (V_{cn} , V_{cp}) track the changes to maintain proper biasing of Mn2 and Mp2.

1.9 Final Design Summary

Table 3: Final design parameters for Simulation 1

Parameter	Value
W	$1.00 \mu\text{m}$
L	$0.18 \mu\text{m}$
M_n	11
M_p	60
$I_{b,main}$	$200 \mu\text{A}$
$ratio_n$	5.8
$ratio_p$	7.5

2 Simulation 2: P- and N-Current Sources

2.1 Objective

Extend the biasing circuit by adding cascode current source transistors (Mn4, Mn5 for NMOS and Mp4, Mp5 for PMOS) to create output branches. Characterize the I-V relationship and output impedance of each branch by sweeping the output voltages V_{on} and V_{op} from 0 to 1.8 V.

2.2 Circuit Configuration

Figure 4 shows the complete circuit with the biasing network and current source branches.

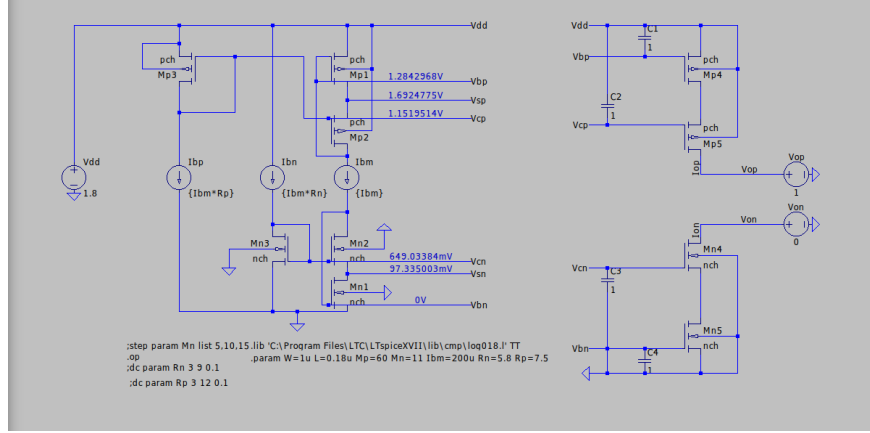


Figure 4: Circuit schematic for Simulation 2: biasing circuit (left) with PMOS and NMOS cascode current sources (right).

The current source branches were added using the same transistor dimensions as the biasing circuit:

- NMOS branch: Mn4 (gate = V_{cn}) and Mn5 (gate = V_{bn})
- PMOS branch: Mp4 (gate = V_{bp}) and Mp5 (gate = V_{cp})
- Decoupling capacitors $C_{big} = 1\text{ F}$ on all bias nodes

2.3 NMOS Current Source Characterization

Figure 5 shows the current $I(V_{on})$ and output impedance $R_{out,n} = 1/\frac{dI}{dV}$ as a function of the output voltage V_{on} .

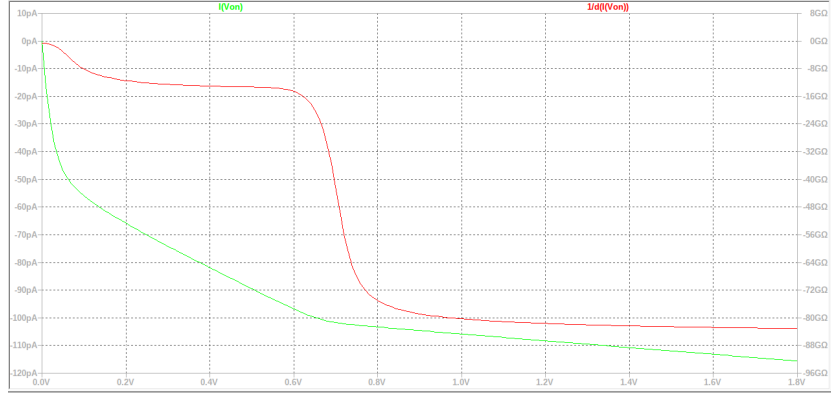


Figure 5: NMOS current source: $I(V_{on})$ (green) and output impedance $1/d(I(V_{on}))$ (red) vs V_{on} .

From the plot, the NMOS cascode current source (Mn4 + Mn5) enters saturation at approximately:

$$V_{on,min} \approx 300 \text{ mV} \quad (3)$$

This represents the minimum voltage headroom required across the two NMOS transistors to maintain saturation.

2.4 PMOS Current Source Characterization

Figure 6 shows the current $I(V_{op})$ and output impedance $R_{out,p} = -1/dI$ as a function of the output voltage V_{op} .

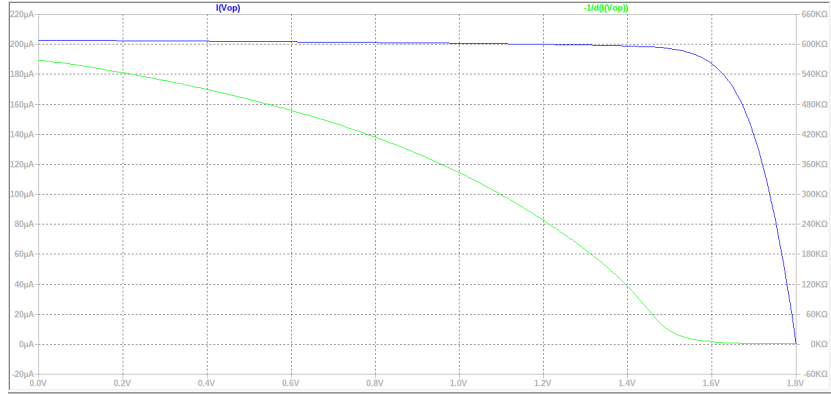


Figure 6: PMOS current source: $I(V_{op})$ (green) and output impedance $-1/d(I(V_{op}))$ (blue) vs V_{op} .

The PMOS branch delivers approximately $200 \mu\text{A}$ in saturation. The tran-

sistors remain saturated for:

$$V_{op} < V_{op,max} \approx 1.5 \text{ V} \quad (4)$$

This means the minimum voltage across Mp4 and Mp5 is:

$$V_{dd} - V_{op,max} = 1.8 - 1.5 = 300 \text{ mV} \quad (5)$$

The output impedance in saturation reaches approximately:

$$R_{out,p} \approx 550 \text{ k}\Omega \quad (6)$$

2.5 Summary of Results

Table 4: Simulation 2 results summary

Parameter	NMOS Branch	PMOS Branch
Saturation current	$\sim 200 \mu\text{A}$	$\sim 200 \mu\text{A}$
Min. voltage for saturation	$\sim 300 \text{ mV}$	$\sim 300 \text{ mV}$
Output impedance R_{out}	$> 1 \text{ M}\Omega$	$\sim 550 \text{ k}\Omega$

The cascode configuration provides high output impedance, which is essential for achieving high gain in amplifier designs.